

Horst Fang

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TECHNICAL SKILLS

Languages: JavaScript/TypeScript, Python, C++, SQL, HTML/CSS
Frontend: React, Next.js, Vite, Tailwind CSS
Backend & APIs: Node.js, FastAPI, LangGraph, Stripe, REST APIs, JWT, API Key Auth
AI / ML: RAG, Vector Search, LangGraph, ChromaDB, OpenAI API, Anthropic API
Databases: PostgreSQL, Supabase, ChromaDB, Prisma ORM
Cloud & DevOps: Docker, GitHub Actions CI/CD, Fly.io, Vercel
Design Tools: Figma, Blender, Adobe Illustrator, Adobe Photoshop, SolidWorks

EXPERIENCE

QuickPOS Technologies Inc.

Jan 2026 – Apr 2026

Software Engineer Intern

Richmond Hill, ON

- Spearheaded development of a **LangGraph RAG agent** with tool-calling and persistent conversation memory via thread-scoped checkpoints across a multi-tenant architecture
- Cut redundant embedding costs by **~70%** with a **delta ingestion pipeline** that skips already-indexed documents and caches embeddings to avoid recomputing unchanged chunks
- Achieved **95%+ test coverage** across all Python services with a CI/CD pipeline via GitHub Actions
- Shipped a React/Vite dashboard and chat UI on Fly.io with a **BFF auth layer** isolating browser from service auth

Ataraxia Apparel Inc.

May 2025 – Dec 2025

Founder

Remote

- Launched a streetwear brand from scratch, developing brand identity, visual language, and go-to-market strategy; grew to **65K+ users** and **\$5K in sales** within one month at a **60% gross margin**
- Led content direction and SEO strategy for atrx.ca; produced and art directed all brand creative across digital channels using Adobe Illustrator and Photoshop
- Managed end-to-end fulfillment and shipping operations, coordinating multi-carrier cross-border logistics and all order and customer management

Trudeau Student Activities Council

May 2024 – June 2025

Head of Publicity

Markham, ON

- Led a 3-member team producing **70+ pieces of digital and physical media**, generating **1M+ views** and a **20% follower increase** using Adobe Illustrator and Photoshop
- Designed and executed weekly events for **1,600+ students**, including a Halloween event with **1,000+ visits** and cumulative revenue of **\$37K+**

PROJECTS

atrx.ca | Next.js, Supabase, PostgreSQL, Prisma ORM, Stripe

May 2026

- Designed and built a custom e-commerce store from scratch in Next.js (App Router) with Supabase, PostgreSQL, Prisma ORM, and Vercel, covering product pages, checkout flow, order state UI, and admin operations
- Implemented **Stripe Checkout Sessions** and **webhook event handling** for payment lifecycle, order state transitions, and fulfillment triggers

Pivot | Swift, SwiftUI, Apple Vision, AVFoundation, WebSocket

May 2026 – Jun 2026

- Built an iOS camera app in SwiftUI with real-time face tracking via **Apple Vision** and a **WebSocket** client (6ms RTT) for servo command routing to an ESP32

Personal Portfolio | Next.js, React Three Fiber, GLSL

Apr 2026

- Built a 3D hologram hero in **React Three Fiber** with a custom **ShaderMaterial** (fresnel, scanlines, cross-hatch, flicker) on Blender-exported .glb models

EDUCATION

University of Waterloo

Expected August 2030

(B.A.Sc.) in Systems Design Engineering

Waterloo, ON

- Recipient of the University of Waterloo President's Scholarship
- **Relevant Coursework:** Digital Computation (C++), Data Structures & Algorithms, Digital Systems